

Beacon Mountain development environment allows developers to write apps for both Intel processor- and ARM-based platforms within the same development environment.

Android's flexible architecture allows most apps to run without any issues on top of a Dalvik Virtual Machine, which abstracts most of the code from the underlying hardware. This allows developers to program apps in Java* without worrying about the differences in functionality on varying hardware architectures.

Many of these Dalvik-based apps that are developed in Java can easily run on Intel devices just like they do on ARM. Moreover, the binary translation functionality makes it easy even for some of the native ARM apps that are dependent on the Android NDK to run on Intel devices without any effort on the part of the developer. The Intel Atom chipsets have innovative features that make them suitable for tablets and mobile phones. Codenamed Silvermont and Haswell, these chipsets are already showing Intel's increasing presence in the new wave of portable devices. While it is impossible to predict the future of the computing industry, trends indicate an impending unification of processor architecture for all devices, regardless of form factor. Intel x86 architecture, both Intel Atom and Intel® Core™ processors, will play a very important in this process.

OVERVIEW

Intel Beacon Mountain, the code name for Intel's Android development environment, provides a full Intel and third-party toolset with IDE integration for all the stages of Android app development. The product not only aims to lower the entry barrier for developing apps for the Android platform on Intel-based devices, it also demonstrates Intel's commitment to Android. The best part about Beacon Mountain is that the tools it installs not only support apps for Intel processor-based devices, but also facilitates development of apps for ARM-based devices, so that developers do not need to jump between development environments to target multiple architectures. Beacon Mountain currently supports Android Jelly Bean 4.1 and later for the Intel-based tools and Android Gingerbread 2.3 and later for third-party tools. The development environment currently supports Intel Core processor-based host systems running Microsoft Windows* 7-, Windows* 8 64-bit-, and Apple OS X*-based machines. Intel plans to support additional platforms in future releases.

The following Intel tools are installed and auto-updated by Beacon Mountain:

- Intel® Hardware Accelerated Execution Manager
- Intel® Graphics Performance Analyzers System Analyzer
- Intel® Integrated Performance Primitives Preview for Android*
- Intel® Threading Building Blocks
- Intel® Software Manager

Standard third-party tools include the Google Android SDK and NDK and the Eclipse* IDE. Intel also provides the Android Design bundles.

You will notice that the Oracle JDK is not included in the bundle; however, Beacon Mountain checks to make sure you have it installed. If you don't, Beacon Mountain points you to the Oracle site to install a copy.

INSTALLATION OF BEACON MOUNTAIN TOOLS

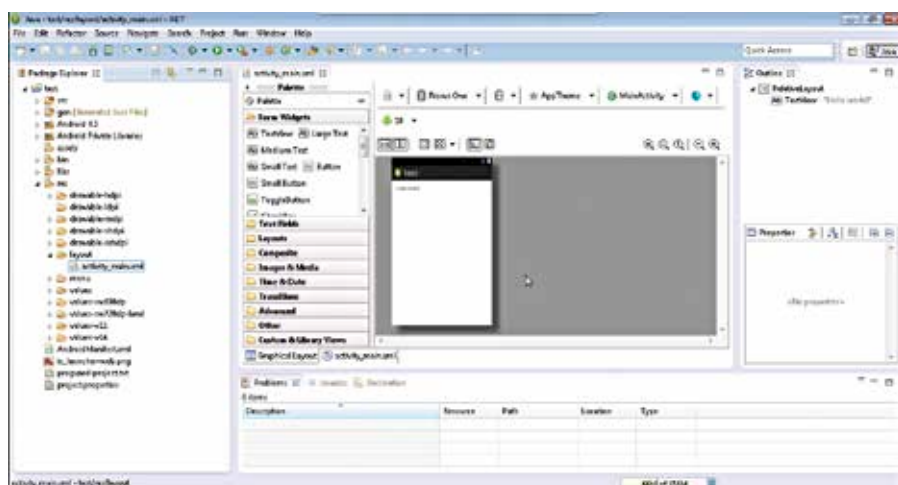
The Beacon Mountain installation binary is available at the Intel web site <http://dgit.in/intelbm>. The binary is basically a web-based installation wizard that downloads all the tools. The installation is as easy as any other one-click installation on a Windows system. The installer shows you the software it will install and asks you to accept licenses in a single view, making the overall installation process very easy. Once the licenses are accepted, the tools are automatically downloaded in under 45 minutes in most cases. The installer automatically detects if the third-party tools are on your system and installs the ones that aren't, so that you can avoid any conflicts. Once installed, the Eclipse IDE shortcut is placed on your desktop.

Alternatively, you can launch Eclipse by going to the Beacon Mountain installation folder. If you are on a Microsoft Windows 7 or Windows 8 system, you can find the 'Beacon Mountain' submenu on the Start menu. Similarly, on a Mac, it may be possible to open the Beacon Mountain folder easily from spotlight. If you want to use the NDK, you can then set its location from the Window > Preferences > NDK submenu. The SDK's locations and all the required components from the Android SDK Manager are already installed by default. Now all you need to do is create an Android Virtual Device.

Uninstallation is equally as easy. In Windows, you can remove any unwanted software elements by going to the Control Panel's Uninstall menu or by running the installation binary and then choosing the 'Remove' option in the wizard.

THIRD-PARTY TOOLS OVERVIEW

The Android Developer Tools (ADT) bundle provides a comprehensive Android development toolkit provided by Google. There are many bundled tools included that supplement the functionality such as the DDMS, Pixel Perfect, Hierarchy Viewer, MonkeyRunner, Traceview, etc. For a more detailed overview of each of these tools, you can take a look at the "Android for Intel Architecture" devworx booklet published in the March 2013 edition of Digit Magazine. It provides an in-depth overview of how to develop Android apps, specifically using third-party and Intel tools. Beacon Mountain officially supports and bundles the Eclipse integrated development environment. Eclipse is one of the most used IDEs with support for the largest number of plug-ins. If you prefer



The Eclipse ADT environment